MAST20009 Vector Calculus

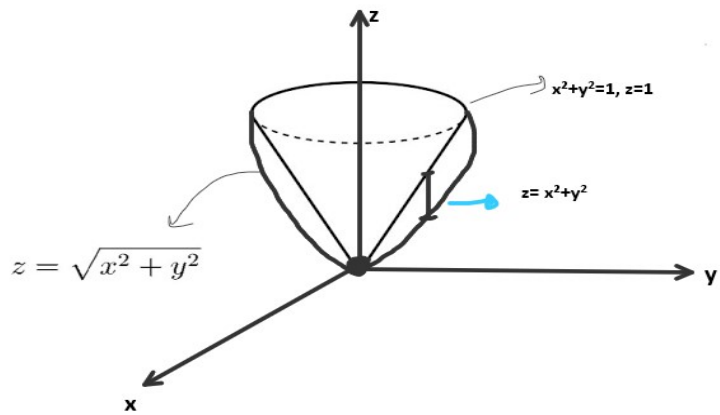
Problem Sheet 4: Double and Triple Integrals

Worked Solutions for selected questions

Q72 (a) Solution:

Region between

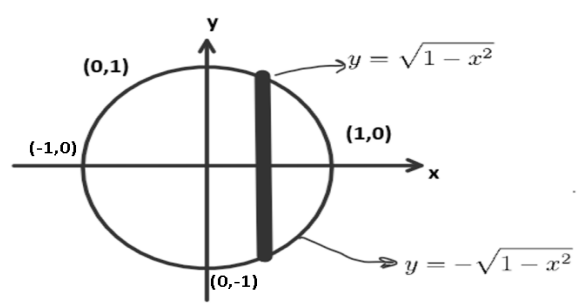
$$z = \sqrt{x^2 + y^2} \quad \text{and } z = x^2 + y^2$$



Bound z first

$$x^2 + y^2 \leq z \leq \sqrt{x^2 + y^2}$$

Project onto xy plane to give $x^2 + y^2 \leq 1$



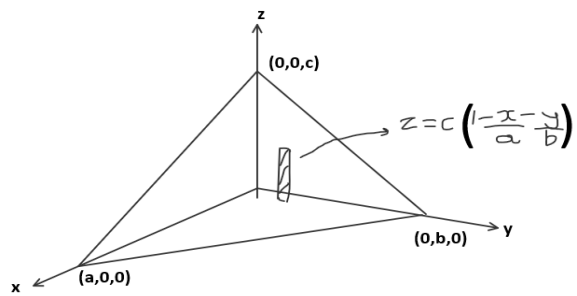
Using vertical strips:

$$-1 \leq x \leq 1, \; -\sqrt{1 - x^2} \leq y \leq \sqrt{1 - x^2}$$

Q73 (a) Volume of region bounded by

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1 \qquad (a,b,c)>0, \quad x=0, y=0, z=0.$$

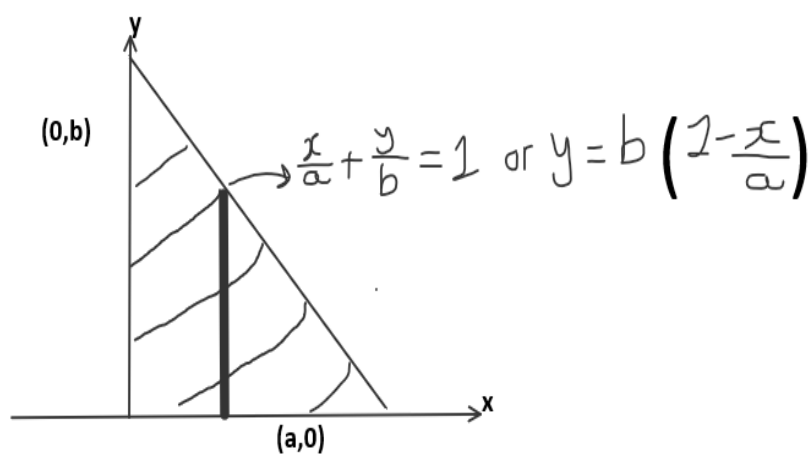
Solution:



Bound z first

$$0 \le z \le c\left(1-\frac{x}{a}-\frac{y}{b}\right)$$

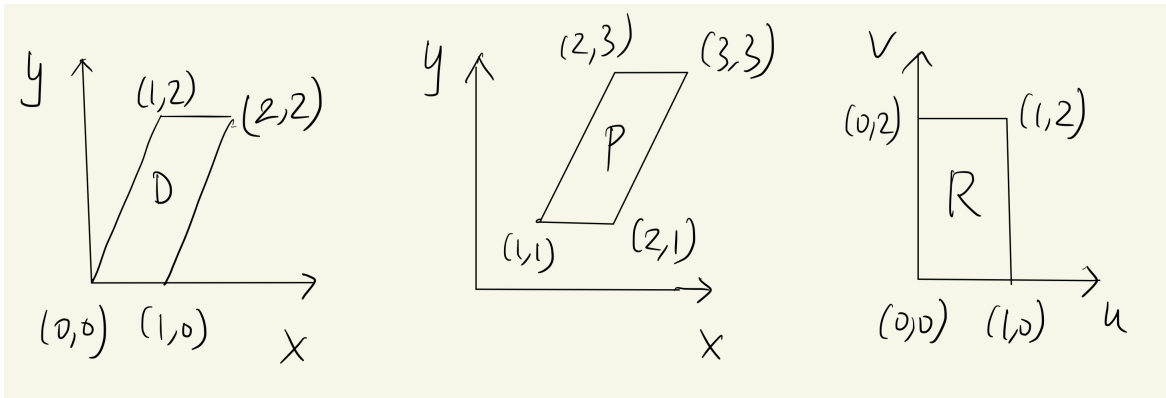
Project onto xy plane to give



$$0 \le x \le a,\ 0 \le y \le b\left(1-\frac{x}{a}\right).$$

$$\begin{aligned}
\text{Volume} &= \int_0^a \int_0^{b\left(1-\frac{x}{a}\right)} \int_0^{c\left(1-\frac{x}{a}-\frac{y}{b}\right)} 1\,dz\,dy\,dx = \int_0^a \int_0^{b\left(1-\frac{x}{a}\right)} c\left(1-\frac{x}{a}-\frac{y}{b}\right)\,dy\,dx \\
&= c\int_0^a \left[y-\frac{xy}{a}-\frac{y^2}{2b}\right]_{y=0}^{y=b\left(1-\frac{x}{a}\right)} dx = c\int_0^a b\left(1-\frac{x}{a}\right)-\frac{b}{a}\left(x-\frac{x^2}{a}\right)-\frac{b}{2}\left(1-\frac{x}{a}\right)^2\,dx \\
&= bc\int_0^a 1-\frac{x}{a}-\frac{x}{a}+\frac{x^2}{a^2}-\frac{1}{2}+\frac{x}{a}-\frac{x^2}{2a^2}\,dx \\
&= bc\int_0^a \frac{1}{2}-\frac{x}{a}+\frac{x^2}{2a^2}\,dx = bc\left[\frac{x}{2}-\frac{x^2}{2a}+\frac{x^3}{6a^2}\right]_{x=0}^{x=a} = bc\left[\frac{a}{2}-\frac{a}{2}+\frac{a}{6}\right] = \frac{abc}{6}.
\end{aligned}$$

Q75



Solution:

(a) Find a linear transformation that maps D (xy plane) to R (uv plane).

Observe the region D is bounded by $y = 0$, $y = 2$, $y = 2x$ and $y = 2x - 2$. The region R is bounded by $u = 0$, $u = 1$, $v = 0$, $v = 2$. So a possible linear transformation is

$$u = \frac{2x - y}{2} \text{ and } v = y.$$

Then

$$x = u + \frac{v}{2}, \ y = v.$$

Alternatively, assume map is given by

$$(0,0) \to (0,0) \tag{1}$$
$$(1,0) \to (1,0) \tag{2}$$
$$(2,2) \to (1,2) \tag{3}$$
$$(1,2) \to (0,2) \tag{4}$$

Let

$$x = au + bv, \quad y = cu + dv$$

(2) gives

$$a = 1, \quad c = 0$$

(3) gives

$$2 = a + 2b, \quad \Rightarrow 2b = 1 \quad \Rightarrow b = \frac{1}{2}$$
$$2 = c + 2d, \quad \Rightarrow 2d = 2 \quad \Rightarrow d = 1$$

(4) gives

$$2b = 1, \quad \Rightarrow b = \frac{1}{2}$$
$$2d = 2 \quad \Rightarrow d = 1$$

Hence a linear transformation is given by

$$T: \quad x = u + \frac{1}{2}v, \quad y = v.$$

(b) Finding

$$\iint_D x + y \, dx \, dy$$

$$\text{Jacobian} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{vmatrix} = 1$$

$$\begin{aligned} \Rightarrow \iint_D x + y \, dx \, dy &= \int_0^1 \int_0^2 \left(u + \frac{3}{2}v \right) dv \, du \\ &= \int_0^1 \left[uv + \frac{3}{4}v^2 \right]_{v=0}^{v=2} du = \int_0^1 2u + 3 \, du \\ &= \left[u^2 + 3u \right]_{u=0}^{u=1} = 4 \, . \end{aligned}$$

(c) Transformation that maps P (xy plane) to R (uv plane).

P is just D translated by 1 in x and y directions.

Hence

$$T: \quad x = u + \frac{1}{2}v + 1, \quad y = v + 1.$$

(d) Find

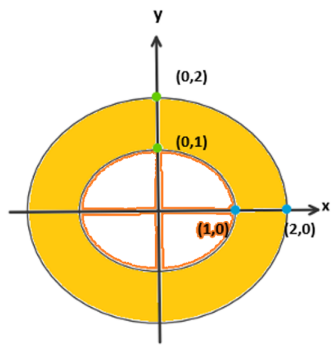
$$\iint_P x + y \, dx \, dy$$

$$\text{Jacobian} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{vmatrix} = 1$$

$$\begin{aligned} \Rightarrow \iint_P x + y \, dx \, dy &= \int_0^1 \int_0^2 \left(u + \frac{3}{2}v + 2 \right) dv \, du \\ &= \int_0^1 \left[uv + \frac{3}{4}v^2 + 2v \right]_{v=0}^{v=2} du = \int_0^1 2u + 3 + 4 \, du = \left[u^2 + 7u \right]_{u=0}^{u=1} = 8 \, . \end{aligned}$$

Q76 (b) Find

$$\iint_R 3x + 8y^2 \, dA, \quad R : 1 \le x^2 + y^2 \le 4$$



Solution:
Use polar coordinates

$$x = r\cos\theta, \quad y = r\sin\theta$$

$$1 \le r \le 2, \quad 0 \le \theta \le 2\pi, \quad \text{Jacobian} = r$$

$$\begin{aligned}
\iint_R 3x + 8y^2 \, dA &= \int_0^{2\pi} \int_1^2 \left(3r\cos\theta + 8r^2\sin^2\theta\right) r \, dr \, d\theta \\
&= \int_0^{2\pi} \int_1^2 3r^2\cos\theta + 8r^3\sin^2\theta \, dr \, d\theta = \int_0^{2\pi} \left[r^3\cos\theta + 2r^4\sin^2\theta\right]_{r=1}^{r=2} d\theta \\
&= \int_0^{2\pi} \left(8\cos\theta + 32\sin^2\theta\right) - \left(\cos\theta + 2\sin^2\theta\right) \, d\theta \\
&= \int_0^{2\pi} 7\cos\theta + 30\sin^2\theta \, d\theta = \int_0^{2\pi} 7\cos\theta + 15 - 15\cos 2\theta \, d\theta \\
&= \left[7\sin\theta + 15\theta - \frac{15}{2}\sin 2\theta\right]_{\theta=0}^{\theta=2\pi} = 30\pi \, .
\end{aligned}$$